

Ride-Sharing Cancellation Analytics – Databricks Lakehouse Project

Business Pitch:

Ride-sharing platforms face significant operational and financial challenges due to frequent ride cancellations. These cancellations lead to revenue loss, poor customer experience, driver inefficiency, and increased wait times. Currently, ride, driver, and customer data are scattered across systems, making it difficult to analyze cancellation patterns, driver reliability, peak failure times, and revenue leakage. Build a Databricks Lakehouse to consolidate all data into one trusted source of truth for actionable insights.

Selected Dataset: Ola & Uber Ride Booking & Cancellation Dataset

Primary Real Dataset Link:

<https://www.kaggle.com/datasets/hetmengar/ola-and-uber-ride-booking-and-cancellation-data?resource=download>

Dataset Summary

- Main rides table: 100,000+ records
- Common columns: Booking_ID, Customer_ID, Driver_ID, Booking_Status (Completed / Cancelled), Pickup_Location, Drop_Location, Booking_Time, Vehicle_Type, Cancellation Reasons (Customer / Driver)
- Strong for cancellation pattern analytics, driver utilization, and revenue analysis

Supplementary Small Master Tables (students create or source small CSVs)

- drivers.csv – 50 to 100 records
- vehicle_types.csv – 5 to 10 records
- cities.csv – 20 to 30 records
- billing.csv – 1,000+ records (synthetic acceptable)
- customer_profiles.csv – 1,000+ records (synthetic acceptable)

Total Suggested Tables

1. rides (real dataset)
2. drivers
3. vehicle_types
4. cities
5. billing
6. customer_profiles

Team Roles (3 Members)

1. Ingestion Engineer – load raw ride and master files, build Bronze Delta tables.
2. Transformation Engineer – clean cancellation flags, fix timestamps, resolve nulls/duplicates, build Silver tables with joins.
3. Analytics Engineer – build Gold KPI layer, dashboards, and final presentation.

Technical Requirements

- Bronze / Silver / Gold medallion architecture
- Delta tables with ACID transactions
- MERGE for incremental daily ride loads
- UPDATE / DELETE for cancellation status corrections
- Schema evolution for new ride attributes
- Time Travel for historical cancellation audits
- SQL + PySpark for transformations and analytics

Execution Steps

Step 1: Ingest rides dataset and all master CSVs into Bronze Delta tables.

Step 2: Clean cancellation_reason values, normalize timestamps, remove nulls and duplicate ride_ids in Silver.

Step 3: Join rides with drivers, vehicle_types, cities, and billing tables in Silver.

Step 4: Use MERGE for new daily ride batch files to handle inserts and updates.

Step 5: Perform Delta operations – UPDATE ride statuses, DELETE test records, enable Time Travel.

Step 6: Build Gold layer aggregations and dashboards.

Gold Layer Outputs

1. Cancellation rate by city and vehicle type
2. Driver reliability and cancellation frequency report
3. Peak hour / daily ride booking and cancellation trends
4. Revenue leakage analysis by cancellation reason
5. Customer behaviour – repeat cancellations and loyalty segments
6. City / zone demand heatmap
7. Average wait time and delay analysis by driver
8. Executive KPI table (total rides, cancellation %, revenue lost, top cities)

Why This Dataset Is Good

- Real-world ride-sharing behavioral data reflecting actual cancellation patterns
- Large enough volume (100,000+ records) to demonstrate scalable lakehouse processing
- Supports meaningful joins with driver, vehicle, and city master tables
- Multiple cancellation reasons enable advanced root-cause analytics

Deliverables

Databricks Notebooks, SQL scripts, PySpark transformation scripts, Bronze/Silver/Gold table screenshots, Gold dashboard visuals, README documentation, final presentation slides.